

Acute Pancreatitis
EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
1. Jin DX, McNabb-Baltar JY, Suleiman SL, et al. Early Abdominal Imaging Remains Over-Utilized in Acute Pancreatitis. Dig Dis Sci. 62(10):2894-2899, 2017 Oct.	Observational-Dx	193 patients	To compare the frequency and evaluate the predictors of early CT/MR utilization for acute pancreatitis (AP) between September 2006–2007 (period A) and September 2014–2015 (period B).	The cohort included 96 AP cases in period A and 97 in period B. There were no significant differences in patient demographics, comorbidity scores, or AP severity. Period B cases manifested decreased rates of the systemic inflammatory response syndrome (SIRS) during the first 24 h of hospitalization (67% period A vs. 43% period B, $p = 0.001$). Independent predictors of early imaging included age ≥ 60 and SIRS or organ failure on day 1. No significant decrease in early CT/MR usage was observed from period A to B on both univariate (49% period A vs. 40% period B, $p = 0.25$) and multivariate (OR 1.0 for period B vs. A, 95% CI 0.5–1.9) analysis.	3
2. Shyu JY, Sainani NI, Sahni VA, et al. Necrotizing pancreatitis: diagnosis, imaging, and intervention. Radiographics. 34(5):1218-39, 2014 Sep-Oct.	Review/Other-Dx	N/A	To present an algorithmic approach to the care of patients with necrotizing pancreatitis and review the use of imaging and interventional techniques in the diagnosis and management of this pathologic condition.	Results not stated in the abstract	4

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3. Banks PA, Bollen TL, Dervenis C, et al. Classification of acute pancreatitis-2012: revision of the Atlanta classification and definitions by international consensus. Gut. 62(1):102-11, 2013 Jan.	Review/Other-Dx	N/A	To present the updated revision of the Atlanta Classification of acute pancreatitis in adults (>18 years).	The revised classification of acute pancreatitis identified two phases of the disease: early and late. Severity is classified as mild, moderate or severe. Mild acute pancreatitis, the most common form, has no organ failure, local or systemic complications and usually resolves in the first week. Moderately severe acute pancreatitis is defined by the presence of transient organ failure, local complications or exacerbation of co-morbid disease. Severe acute pancreatitis is defined by persistent organ failure, that is, organ failure >48 h. Local complications are peripancreatic fluid collections, pancreatic and peripancreatic necrosis (sterile or infected), pseudocyst and walled-off necrosis (sterile or infected). We present a standardised template for reporting CT images.	4
4. Balthazar EJ. Acute pancreatitis: assessment of severity with clinical and CT evaluation. Radiology. 2002;223(3):603-613.	Review/Other-Dx	N/A	Advantages and limitations of the clinical, laboratory, and imaging prognostic indexes are analyzed and discussed.	Contrast-enhanced CT is the imaging modality of choice to help stage the severity of inflammatory processes, detect pancreatic necrosis, and depict local complications. CT has been shown to yield an early overall detection rate of 90% with close to 100% sensitivity after 4 days for pancreatic gland necrosis. The CTSI has shown an excellent correlation with the development of local complications and the incidence of death in this population.	4

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5. Dachs RJ, Sullivan L, Shanmugathan P. Does early ED CT scanning of afebrile patients with first episodes of acute pancreatitis ever change management?. EMERG. RADIOL.. 22(3):239-43, 2015 Jun.	Observational-Dx	248 patients	To examine if CT scanning early in the course of acute pancreatitis (AP) disease (<48 h of symptoms) in afebrile patients with an emergency department (ED) diagnosis of first episode of AP revealed any unanticipated pathology that altered clinical management.	Two hundred forty-eight patients were admitted with an ED diagnosis of AP; 26.2 % (n=65) met inclusion criteria; 70.8 % (n=46) received a CT scan within 48 h of symptom onset. No patient that underwent CT scanning had an unexpected finding (95 % CI, 0.923–1.0). Our results demonstrate that afebrile patients with first episodes of AP do not benefit from early abdominal CT imaging. These results support the ACR Appropriateness Criteria recommendation that CT is not indicated in the first 48 h after symptom onset in unequivocal cases of AP.	3
6. Dobbs NW, Budak MJ, Weir-McCall JR, Vinnicombe SJ, Zealley IA. Acute pancreatitis: a comparison of intervention rates precipitated by early vs guideline CT scan timing. Clin Radiol. 71(10):993-6, 2016 Oct.	Observational-Dx	203 patients	To assess whether computed tomography (CT) examination earlier in acute pancreatitis (AP) precipitates any surgical or radiological intervention.	No intervention was precipitated by performing CT before the sixth day of admission in AP. A statistically significant larger number of interventions were precipitated when CT was performed on the sixth day or later (p<0.05). Of note, this study was conducted using day of admission, rather than day of symptom onset. Six patients underwent repeat CT examination in the same admission after an early CT examination.	3

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7. Bollen TL, Singh VK, Maurer R, et al. A comparative evaluation of radiologic and clinical scoring systems in the early prediction of severity in acute pancreatitis. <i>Am J Gastroenterol</i> . 2012;107(4):612-619.	Observational-Dx	150 patients	To compare the accuracy of CT and clinical scoring systems for predicting the severity of AP on admission.	Of 346 consecutive episodes of AP, there were 159 (46%) episodes in 150 patients (84 men, 66 women; mean age, 54 years; age range, 21-91 years) who were evaluated with a contrast-enhanced CT scan (n=131 episodes) or an unenhanced CT scan (n=28 episodes) on the first day of admission. Clinically severe AP was diagnosed in 29/159 (18%) episodes; 9 (6%) patients died. Overall, the Balthazar grading system (any CT technique) and CTSI (contrast-enhanced CT only) demonstrated the highest accuracy among the CT scoring systems for predicting severity, but this was not statistically significant. There were no statistically significant differences between the predictive accuracies of CT and clinical scoring systems.	3
8. Shinagare AB, Ip IK, Raja AS, Sahni VA, Banks P, Khorasani R. Use of CT and MRI in emergency department patients with acute pancreatitis. <i>Abdom Imaging</i> . 40(2):272-7, 2015 Feb.	Observational-Dx	101 patients	To assess the utility of CT and MRI in patients with acute pancreatitis (AP) presenting to emergency department (ED)	Of 101 patients admitted with AP (60 women, 41 men; mean age 52 years, range 20–89), 63 (62.4%) underwent imaging; only one (1.6%) showed pancreatic necrosis. 88 (87.1%) patients could have been clinically diagnosed without imaging based on presence of abdominal pain and elevated laboratory values; 13 (12.9%) required imaging for diagnosis. Of 88 patients who met AP diagnostic criteria without imaging, 50 (56.8%) nonetheless underwent imaging, with AP without necrosis seen in 34 (68.0%), pancreatic necrosis in one (2.0%), sequelae of prior AP in four (8.0%), and no abnormality in 11 (22.0%).	3

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9. Mofidi R, Lee AC, Madhavan KK, Garden OJ, Parks RW. The selective use of magnetic resonance cholangiopancreatography in the imaging of the axial biliary tree in patients with acute gallstone pancreatitis. <i>Pancreatology</i> . 2008;8(1):55-60.	Observational-Dx	249 patients	To assess the impact following the introduction of MRCP in the management of acute gallstone pancreatitis in a tertiary referral unit.	There was no significant difference in serum bilirubin levels [ERCP 43 mmol/l (18-204) vs MRCP 39 mmol/l (24-180), P=NS] or the proportion of patients with CBD stones [ERCP 10 (17.5%) vs MRCP 7 (14.2%), P=NS] between the two groups. Patients who underwent MRCP had a shorter median hospital stay [MRCP 5 days (range: 3-14) vs ERCP 9 days (range: 4-20), P<0.01] and higher rate of cholecystectomy during the index admission (MRCP 83.3% vs ERCP 67.2%, P<0.05). There was a high degree of correlation between preoperative MRCP results and findings of subsequent intraoperative cholangiogram or therapeutic ERCP (area under ROC curve: 0.94).	3
10. van Santvoort HC, Bakker OJ, Besselink MG, et al. Prediction of common bile duct stones in the earliest stages of acute biliary pancreatitis. <i>Endoscopy</i> . 43(1):8-13, 2011 Jan.	Observational-Dx	167 patients	To evaluate commonly used biochemical and radiological predictors of common bile duct (CBD) stones in a large prospective cohort of patients with acute biliary pancreatitis who were undergoing early endoscopic retrograde cholangiopancreatography (ERCP).	Out of 167 patients, 94 (56 %) had predicted severe acute biliary pancreatitis, 51 (31 %) exhibited a dilated CBD and 15 (9 %) had CBD stones on ultrasonography and/or CT. ERCP was performed at a median of 0 days (interquartile range 0–1) after admission. CBD stones were found during ERCP in 89/167 patients (53 %). In univariate analysis, the only parameters significantly associated with CBD stones were GGT (per 10 units increase: odds ratio 1.02, 95% CI 1.01–1.03, P = 0.001) and alkaline phosphatase (per 10 units increase: odds ratio 1.03, 95% CI 1.00–1.05, P = 0.028). These and all other tested parameters, however, showed poor positive predictive value (ranging from 0.53 to 0.69) and poor negative predictive value (ranging from 0.46 to 0.67).	3

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11. Cavdar F, Yildar M, Tellioglu G, Kara M, Tilki M, Titiz MI. Controversial issues in biliary pancreatitis: when should we perform MRCP and ERCP?. Pancreatolgy. 14(5):411-4, 2014 Sep-Oct.	Observational-Dx	60 patients	To assess the number and timing of spontaneous passage of BDS using magnetic resonance cholangiopancreatography (MRCP) and to determine the effect of this approach on endoscopic retrograde cholangiopancreatography (ERCP).	MRCP revealed choledocholithiasis in 20 (33%) of the 60 patients. In the control MRCP imaging, choledocholithiasis was detected in 16 of 20 (80% of those who had stone initially) patients. ERCP was performed in these patients and in 2 patients who did not have BDS on the control MRCP but whose bilirubin values and cholestatic enzyme levels had not decreased. ERCP verified choledocholithiasis in 16 of the 18 patients. The positive predictive value of MRCP was 93.7% (15/16).	3
12. Cucher D, Kulvatunyou N, Green DJ, Jie T, Ong ES. Gallstone pancreatitis: a review. Surg Clin North Am. 94(2):257-80, 2014 Apr.	Review/Other-Dx	N/A	To review gallstone pancreatitis (GSP) with a focus on surgical management, including the appropriate timing and choice of interventions.	Results not stated in the abstract	4
13. Anand G, Patel YA, Yeh HC, et al. Factors and Outcomes Associated with MRCP Use prior to ERCP in Patients at High Risk for Choledocholithiasis. Can J Gastroenterol Hepatol. 2016;5132052, 2016.	Observational-Dx	2313 patients	To evaluate factors and outcomes associated with performing magnetic resonance cholangiopancreatography (MRCP) prior to endoscopic retrograde cholangiopancreatography (ERCP) among patients at high risk for choledocholithiasis.	224 patients classified as high risk, of whom 176 (79%) underwent ERCP only, while 48 (21%) underwent MRCP prior to ERCP. Patients undergoing MRCP experienced longer time to ERCP (72 hours versus 35 hours, ?? < 0.0001), longer length of stay (8 days versus 6 days, ?? = 0.02), higher hospital charges (\$23,488 versus \$19,260, ?? = 0.08), and higher radiology charges (\$3,385 versus \$1,711, ?? < 0.0001). The presence of common bile duct stone(s) on ultrasound was the only independent factor associated with less use of MRCP (OR 0.09, ?? < 0.0001).	3

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14. Cho JH, Kim TN, Kim SB. Comparison of clinical course and outcome of acute pancreatitis according to the two main etiologies: alcohol and gallstone. BMC Gastroenterol. 15:87, 2015 Jul 25.	Observational-Dx	126 patients	To investigate the difference between the clinical course of alcoholic and biliary acute pancreatitis (AP)	Hemoglobin, hematocrit, and serum C-reactive protein level measured after admission for 24 h were significantly higher in the alcohol group than in the biliary group. Incidence of pseudocyst formation was significantly higher in the alcohol group than in the biliary group (20.0 % vs. 6.6 %, P = 0.023). Among prognostic scoring systems, only CTSI showed significant difference (P < 0.001) with a mean score of 3.0 ± 0.9 in the alcohol group and 1.7 ± 1.2 in the biliary group. Severe AP with organ failure persisting beyond 48 h was observed in 12 patients (24.0 %) in the alcohol group and one patient (1.3 %) in the biliary group (P < 0.001). There were 4 mortalities in the alcohol group only (P = 0.012).	3
15. Turkvatan A, Erden A, Turkoglu MA, Secil M, Yener O. Imaging of acute pancreatitis and its complications. Part 1: acute pancreatitis. [Review]. Diagn Interv Imaging. 96(2):151-60, 2015 Feb.	Review/Other-Dx	N/A	To present an overview of the acute pancreatitis, clarify confusing terminology, underline the role of ultrasound, computed tomography and magnetic resonance imaging according to the proper clinical context and compare the advantages and limitations of each modality.	Results not stated in abstract	4
16. Wu BU, Conwell DL. Acute pancreatitis part I: approach to early management. Clin Gastroenterol Hepatol. 2010;8(5):410-416, quiz e456-418.	Review/Other-Dx	1 patient	Review diagnosis and management of AP.	No results stated in abstract.	4
17. Manikkavasakar S, AlObaidy M, Busireddy KK, et al. Magnetic resonance imaging of pancreatitis: an update. [Review]. World Journal of Gastroenterology. 20(40):14760-77, 2014 Oct 28. World J Gastroenterol. 20(40):14760-77, 2014 Oct 28.	Review/Other-Dx	N/A	To address new trends in clinical pancreatic MR imaging emphasizing its role in imaging all types of acute and chronic pancreatitis, pancreatitis complications and other important differential diagnoses that mimic pancreatitis.	Results not stated in the abstract	4

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18. Zhao K, Adam SZ, Keswani RN, Horowitz JM, Miller FH. Acute Pancreatitis: Revised Atlanta Classification and the Role of Cross-Sectional Imaging. [Review]. AJR Am J Roentgenol. 205(1):W32-41, 2015 Jul.	Review/Other-Dx	N/A	To review the imaging findings associated with acute pancreatitis and its complications on cross-sectional imaging and discusses the role of imaging in light of this revision.	Results not stated in the abstract	4
19. Aydin H, Tatar IG, Hekimoglu B. The role of diffusion weighted MR imaging in the diagnosis of acute pancreatitis. Int J Emerg Ment Health. 16(2):308-14, 2014.	Review/Other-Dx	N/A	To understand the utility of Diffusion-weighted MR imaging (DWI) in the diagnosis of acute pancreatitis.	Results not stated in the abstract	4
20. Hocaoglu E, Aksoy S, Akarsu C, Kones O, Inci E, Alis H. Evaluation of diffusion-weighted MR imaging in the diagnosis of mild acute pancreatitis. Clin Imaging. 39(3):463-7, 2015 May-Jun.	Observational-Dx	22 healthy subjects and 40 patients	To determine the diagnostic value of diffusion-weighted magnetic resonance imaging in the identification of acute mild pancreatitis with low Ranson scores	There was a significant reduction in mean pancreatic apparent diffusion coefficient among the acute pancreatitis patients ($1.46 \pm 2.80 \times 10^{-3} \text{ mm}^2/\text{s}$) relative to the healthy subjects ($1.69 \pm 2.26 \times 10^{-3} \text{ mm}^2/\text{s}$).	2
21. Zaheer A, Singh VK, Qureshi RO, Fishman EK. The revised Atlanta classification for acute pancreatitis: updates in imaging terminology and guidelines. Abdom Imaging. 38(1):125-36, 2013 Feb.	Review/Other-Dx	N/A	To discuss the radiographic evolution of Acute pancreatitis (AP) as well as the associated complications with special emphasis on the terminology and the role of imaging in different phases of the disease.	Results not stated in the abstract	4

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22. Peng R, Zhang XM, Ji YF, et al. Pancreatic duct patterns in acute pancreatitis: a MRI study. PLoS ONE. 8(8):e72792, 2013.	Observational-Dx	239 patients with acute pancreatitis and 125 controls	To study the MRI findings of the pancreatic duct in patients with acute pancreatitis.	The frequency of MPD segment visualization in the control group was higher than that in the acute pancreatitis group (p,0.05). The number of MPD segments visualized was negatively correlated with the MRSI score (p,0.05) and the APACHE II score (p,0.05). There was no difference in the MPD diameter between the acute pancreatitis and control groups or among the patients with different severities of acute pancreatitis (p,0.05). The prevalence of pancreatic duct disruption was 7.9% in the acute pancreatitis group. The prevalences of pancreatic duct disruption were 4.8% and 15.3% in the mild and severe acute pancreatitis groups based on the APACHE II score, respectively, and were 0%, 5.7% and 43.5% in the mild, moderate and severe acute pancreatitis groups according to the MRSI score, respectively. The prevalence of pancreatic duct disruption was correlated with the severity of acute pancreatitis based on the APACHE II score (p,0.05) and MRSI score (p,0.05).	2

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23. Ji YF, Zhang XM, Li XH, et al. Gallbladder patterns in acute pancreatitis: an MRI study. Acad Radiol. 19(5):571-8, 2012 May.	Observational-Dx	197 patients	To assess the gallbladder patterns on magnetic resonance imaging (MRI) associated with acute pancreatitis (AP).	Of the 197 patients with AP, 81% were classified as edematous and 19% as necrotizing on MRI. There were 35%, 59%, and 6% of patients with mild, moderate, and severe AP according to the magnetic resonance severity index, respectively. Seventy-six percent of patients had at least one gallbladder abnormality on MRI, including a thickened gallbladder wall (42%), pericholecystic fluid (38%), gallbladder stones (35%), an enlarged gallbladder (24%), dilatation of the common bile duct (16%), and subserosal edema (15%). Eighty-nine percent of patients (34 of 38) with necrotizing AP had gallbladder abnormalities, which was significantly higher than the 72% of patients (115 of 159) with edematous AP ($P < .05$). The prevalence of gallbladder abnormalities was 64% in patients with mild AP, 81% in those with moderate AP, and 91% in those with severe AP ($P < .05$ among the three groups).	2

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24. Li XH, Zhang XM, Ji YF, et al. Renal and perirenal space involvement in acute pancreatitis: An MRI study. Eur J Radiol. 81(8):e880-7, 2012 Aug.	Observational-Dx	115 patients	To study the prevalence and characteristics of renal and perirenal space involvement and its relation to the severity of acute pancreatitis (AP) using MRI.	In the 115 patients with AP, the renal and perirenal space abnormalities detected included renal parenchymal abnormalities (0.8%), abnormalities of the renal collecting system (2.6%), renal vascular abnormalities (1.7%), thickened renal fascia (99%), perirenal stranding (62%) and perirenal fluid collection(40%). The prevalence of perirenal space abnormalities was correlated with the severity of AP based on MRSI ($P < 0.05$). The renal ADC values were lower in patients with abnormal kidney function than in those without kidney injury ($P < 0.05$). The prevalence of kidney function abnormalities was 9.4%, 32% and 100% in mild, moderate, and severe AP cases, respectively ($P = 0.00$).	2
25. Bakker OJ, van Santvoort H, Besselink MG, et al. Extrapaneatric necrosis without pancreatic parenchymal necrosis: a separate entity in necrotising pancreatitis?. Gut. 62(10):1475-80, 2013 Oct.	Observational-Dx	639 patients	To investigate whether the rate of complications and mortality of patients with extrapancreatic necrosis only (EXPN) differs from that of patients with pancreatic necrosis with or without extrapancreatic necrosis. To determine whether the rates of complications differ between patients with EXPN or pancreatic necrosis who develop infected necrosis.	315 patients with EXPN were compared with 324 patients with pancreatic parenchymal necrosis. Patients with EXPN less often suffered from complications: persistent organ failure (21% vs 45%, $p < 0.001$), persistent multiple organ failure (15% vs 36%, $p < 0.001$), infected necrosis (16% vs 47%, $p < 0.001$), intervention (18% vs 57%, $p < 0.001$) and mortality (9% vs 20%, $p < 0.001$). When infection of extrapancreatic necrosis developed, outcomes between groups were equal (mortality with infected necrosis: EXPN 28% vs pancreatic necrosis 18%, $p = 0.16$).	1

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26. Kamal A, Singh VK, Akshintala VS, et al. CT and MRI assessment of symptomatic organized pancreatic fluid collections and pancreatic duct disruption: an interreader variability study using the revised Atlanta classification 2012. <i>Abdom Imaging</i> . 40(6):1608-16, 2015 Aug.	Observational-Dx	29 patients	To compare CT and MRI for fluid/debris component estimate and pancreatic duct (PD) communication with organized pancreatic fluid collections in acute pancreatitis. To evaluate fat density globules on CT as marker for debris.	Necrotizing pancreatitis seen on CT in 27 (93%, k 0.119) vs. 20 (69%, k 0.748) patients on MRI. CT identified 42 WON and 4 pseudocysts vs. 34 WON, and 12 pseudocysts on MRI. Higher interreader agreement for percentage fluid volume on MRI (k = 0.55) vs. CT (k = 0.196). Accuracy of CT in evaluation of percentage fluid volume was 65% using T2WI MRI used as standard. Fat globules identified on CT in 13(65%) out of 20 collections containing <75% fluid vs. 4(15%) out of 26 collections containing >75% fluid (p = 0.0001). PD involvement confidently excluded on CT in 68% collections vs. 93% on MRI.	2
27. Sternby H, Verdonk RC, Aguilar G, et al. Significant inter-observer variation in the diagnosis of extrapancreatic necrosis and type of pancreatic collections in acute pancreatitis - An international multicenter evaluation of the revised Atlanta classification. <i>Pancreatology</i> . 16(5):791-7, 2016 Sep-Oct.	Observational-Dx	285 patients	To analyse the interobserver agreement among radiologists in evaluating CT morphology by using the new revised Atlanta classification (RAC) criteria in patients with acute pancreatitis (AP).	In total, 285 patients with 388 CTs were included. For most CT criteria, interobserver agreement was moderate to substantial. In four categories, the center independent kappa values were fair: extrapancreatic necrosis (EXPN) (0.326), type of pancreatitis (0.370), characteristics of collections (0.408), and appropriate term of collections (0.356). The fair kappa values relate to discrepancies in the identification of extrapancreatic necrotic material. The local radiologists diagnosed EXPN (33% versus 59%, P < 0.0001) and non-homogeneous collections (35% versus 66%, P < 0.0001) significantly less frequent than the central expert. Cases read by the central expert showed superior correlation with clinical outcome.	2

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28. Cai DM, Parajuly SS, Ling WW, Li YZ, Luo Y. Diagnostic value of contrast enhanced ultrasound for splenic artery complications following acute pancreatitis. <i>World J Gastroenterol.</i> 20(4):1088-94, 2014 Jan 28.	Observational-Dx	118 patients	To assess the value of contrast-enhanced ultrasound (CEUS) in diagnosing splenic artery complications (SACs) after acute pancreatitis (AP).	Nine patients were diagnosed as having SACs after AP by CECT among the 118 patients. The patients with SACs were diagnosed with severe acute pancreatitis (SAP). Among them, 6 lesions were diagnosed as splenic artery embolism, 5 as splenic artery aneurysms, and 1 as splenic arterial stenosis. No lesion was diagnosed by pancreatic CEUS and 5 lesions were diagnosed by splenic CEUS. By splenic CEUS, 4 cases were diagnosed as splenic artery embolism and 1 as splenic arterial stenosis. The accuracy of splenic CEUS in diagnosis of SACs in SAP was 41.7% (5/12), which was higher than that of pancreatic CEUS (0%).	3
29. Islim F, Salik AE, Bayramoglu S, Guven K, Alis H, Turhan AN. Non-invasive detection of infection in acute pancreatic and acute necrotic collections with diffusion-weighted magnetic resonance imaging: preliminary findings. <i>Abdominal Imaging.</i> 39(3):472-81, 2014 Jun.	Observational-Dx	20 patients	To evaluate the contribution of diffusion-weighted magnetic resonance imaging (DW-MRI) to the detection of infection in acute pancreatitis-related collections.	Apart from one false positive result, the presence of infection was detected by DW-MRI with 95.2% accuracy. The sensitivity and accuracy of DW-MRI were higher than CT for the detection of infection. The ADC values in the central parts of the collections were significantly different between the infected and sterile groups.	1

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30. Rana SS, Sharma V, Sharma R, Gupta R, Bhasin DK. Endoscopic ultrasound guided transmural drainage of walled off pancreatic necrosis using a "step - up" approach: A single centre experience. <i>Pancreatology</i> . 17(2):203-208, 2017 Mar - Apr.	Observational-Dx	86 patients	To evaluate the safety and efficacy of "endoscopic step up approach" comprising of initial drainage with multiple plastic stents and/or nasocystic catheter (NCC) followed by fully covered self-expanding metallic stents (FCSEMS) and/or direct endoscopic necrosectomy (DEN) in non-responders for the treatment of walled off necrosis (WON).	Patients presented 10.8 ± 2.8 weeks after an acute attack of acute pancreatitis and mean size of WON was 10.7 ± 2.9 cm. EUS guided transmural drainage was technically successful in 85/86 (98.8%) patients and 70 (82.4%) were drained with multiple 7/10Fr plastic stents alone while DEN was needed in 9 (10.6%) and FCSEMS was inserted in 6 (7%) patients. All patients had successful outcome with none requiring surgery. The patients who needed DEN/FCSEMS presented earlier and had large size collection with more solid necrotic debris as compared to patients treated with multiple plastic stents alone. The complications were pneumoperitoneum (n = 1), bile leak following cholecystectomy (n = 1), development of external pancreatic fistula following percutaneous drainage (n = 1) and gastrointestinal bleed (n = 1).	3
31. Dhaka N, Samanta J, Kochhar S, et al. Pancreatic fluid collections: What is the ideal imaging technique?. [Review]. <i>World J Gastroenterol</i> . 21(48):13403-10, 2015 Dec 28.	Review/Other-Dx	N/A	To review the role of all available imaging modalities in differentiating Pancreatic fluid collections (PFCs) in patients of Acute pancreatitis (AP).	Results not stated in the abstract	4

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32. Jang JW, Kim MH, Oh D, et al. Factors and outcomes associated with pancreatic duct disruption in patients with acute necrotizing pancreatitis. <i>Pancreatology</i> . 16(6):958-965, 2016 Nov - Dec. <i>Pancreatology</i> . 16(6):958-965, 2016 Nov - Dec.	Observational-Dx	84 patients	To evaluate the predictors, clinical consequences, and outcomes of concomitant MPD disruption/leakage in patients with ANP.	MPD disruption was documented in 38% (32/84) of the ANP patients. Extensive necrosis, enlarging/refractory pancreatic fluid collections (PFCs), persistence of amylase-rich output from percutaneous drainage, and amylase-rich ascites/pleural effusion were more frequently associated with MPD disruption. Hospital stay was prolonged (mean 55 vs. 29 days) and recurrence of PFCs (41% vs. 14%) was more frequent in the MPD disruption group, although mortality did not differ between ANP patients with and without MPD disruption. Subgroup analysis between complete disruption (n = 14) and partial disruption (n = 18) revealed a more frequent association of extensive necrosis and full-thickness glandular necrosis with complete disruption. The success rate of endoscopic transpapillary pancreatic stenting across the stricture site was lower in complete disruption (20% vs. 92%). Patients with complete MPD disruption also showed a high rate of PFC recurrence (71% vs. 17%) and required surgery more often (43% vs. 6%).	3
33. American College of Radiology. ACR Appropriateness Criteria® Radiation Dose Assessment Introduction. Available at: https://edge.sitecorecloud.io/american-coldf5f-acrorgf92a-productioncb02-3650/media/ACR/Files/Clinical/Appropriateness-Criteria/ACR-Appropriateness-Criteria-Radiation-Dose-Assessment-Introduction.pdf .	Review/Other-Dx	N/A	To provide evidence-based guidelines on exposure of patients to ionizing radiation.	No abstract available.	4

Evidence Table Key

Study Quality Category Definitions

- *Category 1* The study is well-designed and accounts for common biases.
- *Category 2* The study is moderately well-designed and accounts for most common biases.
- *Category 3* There are important study design limitations.
- *Category 4* The study is not useful as primary evidence. The article may not be a clinical study or the study design is invalid, or conclusions are based on expert consensus. For example:
 - a. The study does not meet the criteria for or is not a hypothesis-based clinical study (e.g., a book chapter or case report or case series description);
 - b. The study may synthesize and draw conclusions about several studies such as a literature review article or book chapter but is not primary evidence;
 - c. The study is an expert opinion or consensus document.
- Meta-analysis
 - a. *Good quality* – the study design, methods, analysis, and results are valid and the conclusion is supported.
 - b. *Inadequate quality* – the study design, analysis, and results lack the methodological rigor to be considered a good meta-analysis study.

Abbreviations Key

Dx = Diagnostic
Tx = Treatment